Chapter 4 lecture script, Lecture 1: Maximum likelihood estimation, the method

Draft for the course author's review. **Target: 150 minutes in total, with 127 minutes of slides on 33 slides (including the discussion, the two practice questions and one live demo of 18 minutes), 3 minutes of slack and two breaks of 10 minutes.** The breaks separate the three blocks of the lecture. The only code that is run live is the demo of slide 31, from the script `demo-profile.R`.

This script follows the first part of the Chapter 4 theory, up to the section "Parameterisation, slices and profiles", with a short introduction to the first part of the R practical. Lecture 2 (`../Lecture_2/lecture-script.md`) covers the second part of the chapter: uncertainty, the properties of maximum likelihood estimators and model comparison. The slides carry keywords only; the narrative is here.

Slide 3 uses your diagram `ModelCycle.png` from the Chapter 1 lecture folder. Slide 16 uses the schematic `optimiser-search.svg` and slide 18 the schematic `fit-pipeline.svg`. Slide 7 uses two figures made for the lecture, `poisson-mass.png` and `fir-likelihood.png`, which have no counterpart in the chapter. Slides 15 (`likelihood-curve.png`), 26 (`owl-fits.png`), 27 (`likelihood-surface.png`) and 28 and 30 (`slice-profile.png`, the same figure twice) use copies of the chapter figures `fig-likelihood_curve`, `fig-owl_fits`, `fig-likelihood_surface` and `fig-slice_profile`. All are in `lecture-assets/`, and `lecture-assets/figures.R` makes them, the copies from the code of the chapter. If the chapter figures change, run that script again.

Timing and slide sequence

Times are measured from the start of the session and include the breaks.

Slide	Topic	Minutes	Elapsed time
1	Ecological Models and Data Analysis	1	0–1
2	Overview of the lecture	2	1–3
3	Where we are: the modelling cycle	1	3–4
4	From the method of moments to maximum likelihood	2	4–6
5	The likelihood function	4	6–10
6	Probability and likelihood	3	10–13

Slide	Topic	Minutes	Elapsed time
7	Probability mass and likelihood curve	2	13–15
8	A likelihood by hand	4	15–19
9	A likelihood is not the probability of a parameter value	4	19–23
10	Discussion: probability or likelihood?	3	23–26
11	The log-likelihood	4	26–30
12	Log-likelihoods in R	3	30–33
13	Practice 1: crab burrows	5	33–38
	Break	10	38–48
14	The maximum likelihood estimator	5	48–53
15	The likelihood curve	4	53–57
16	Numerical optimisation	3	57–60
17	The negative log-likelihood in R	4	60–64
18	From the negative log-likelihood to a fit	3	64–67
19	A fit in R	5	67–72
20	The result of `optim()`	3	72–75
21	Constraints and transformations	4	75–79
22	Invariance	4	79–83
23	Practice 2: back to the ecological scale	3	83–86

Slide	Topic	Minutes	Elapsed time
24	Models with several parameters	4	86–90
25	A fit with two parameters in R	5	90–95
	Break	10	95–105
26	Checking the fitted model	3	105–108
27	The likelihood surface	4	108–112
28	Parameterisation	4	112–116
29	Slices and profiles	3	116–119
30	Slice against profile	4	119–123
31	Live demo: a likelihood profile in R	2 + 16	123–141
32	The Chapter 4 R practical: first part	2	141–143
33	Summary and closing questions	4	143–147
	Slack for discussion	3	147–150
	Total	**150**	

Slides 127 minutes (blocks of 38, 47 and 42, the demo included), slack 3, breaks 20: 150 minutes.

Slides 1 and 2 — Ecological Models and Data Analysis; Overview of the lecture

In this lecture, we learn how to estimate the parameters of a model by maximum likelihood. In Chapter 3 we estimated parameters with the method of moments, which only uses the mean and the variance of a sample, and we described the uncertainty of an estimate with a sampling distribution, a standard error and a confidence interval. Maximum likelihood uses the probability that the model assigns to every observation, so it applies to any model for which we can write a probability mass or density function. That is why it is the method of estimation in the rest of the course.

The chapter has two parts and each has its own session. This session builds the method: the likelihood function, the estimate that maximises it, the way to find that estimate with an optimiser, and what changes when a model has several parameters. The next session asks what we can say once a model is fitted: how uncertain the estimates are, why the method works, and how to compare models.

The session has three blocks, separated by two breaks. The first block is about the likelihood and the log-likelihood and ends with the first practice question. The second is about estimation: the maximum likelihood estimator, the fit in R with `optim()`, and the transformations that keep parameters within their range. The third is about models with two parameters, the shape of the likelihood surface, slices and profiles, and it contains the live demo.

Delivery

Take the overview at the pace of the list: read each keyword and its question, and say in one sentence that the questions are answered in that order. Do not explain any of the terms yet.

Slide 3 — Where we are: the modelling cycle

On the diagram of the modelling cycle, this lecture is the "likelihood (4)" and "optimise (4)" entries of the estimation box. Chapter 2 gave us the distributions of the model box and Chapter 3 the first tools of the estimation box, with confidence intervals (3). The likelihood is the quantity from which the rest of the estimation box, including the confidence intervals of Chapter 4, is calculated.

Delivery

One minute. Point to "likelihood (4)" and "optimise (4)" only; do not walk through the cycle again.

Slide 4 — From the method of moments to maximum likelihood

The method of moments uses two summaries of the sample, its mean and its variance, and chooses the parameters so that the model reproduces them. Maximum likelihood uses more: the probability that the model assigns to every single observation. Its only requirement is that we can write a probability mass function or a density function for the observations, so it works for any model of the course.

Everything that we learnt in Chapter 3 remains true. An estimator is a rule applied to a sample, an estimate is the value that it returns for the data that we have, and its uncertainty is described by its sampling distribution, its standard error and a confidence interval. The only thing that changes is the estimator, so the new part of the chapter is how the estimator is defined and calculated.

Two examples run through the chapter. The first is the count of balsam fir seedlings in 50 quadrats of the Chapter 3 lecture, with a Poisson model. The second is the number of calls made by barn owl nestlings at 122 visits of a parent to a nest, with a Negative Binomial model that has two parameters. The Poisson example is simple enough for calculations by hand, and the owls show what changes when a model has two parameters.

Slide 5 — The likelihood function

The **likelihood** of a sample is the joint probability of the observations, read as a function of the parameters with the data held fixed. For observations $x_1, ..., x_n$ and a parameter θ (as in Chapter 3, θ stands for a parameter in general), it is $L(\theta \mid x_1, ..., x_n) = P(X_1 = x_1, ..., X_n = x_n \mid \theta)$.

The vertical bar reads "for a given value of": $P(X = x \mid \theta)$ is the probability of x for a given value of θ, and $L(\theta \mid x_1, ..., x_n)$ is the likelihood of θ for a given sample. The same symbol is used for conditional distributions, which Chapter 7 introduces. The second equality of the formula

holds because the observations are independent, so their joint probability is the product of their individual probabilities. That is the assumption that Chapter 2 announced would be fundamental for estimation, in the section on joint distributions and independence. For continuous data each probability mass is replaced by the density of the observation.

The example is the count of seedlings in each quadrat, $X_i \sim \text{Poisson}\,(\lambda)$. The data are the counts of balsam fir seedlings in 50 quadrats of five by five feet, laid out in a grid of five rows of ten quadrats (Davison and Hinkley 1997, *Bootstrap Methods and Their Application*, Cambridge University Press), which come with R as the data set `fir` of the package `boot`. We treat the counts as i.i.d. Neighbouring quadrats of a grid may be more alike than distant ones, so this is a simplification. The course does not model this spatial dependence; Chapter 7 covers a related case, observations that come in groups.

Optional remark (history)

Fisher introduced the method in 1912, in a paper written as an undergraduate student, and called it an "absolute criterion". He introduced the word likelihood in 1921, to keep it apart from a probability, and the name maximum likelihood in 1922. The idea of choosing the value under which the observations are most probable goes back to Lambert, Daniel Bernoulli and Gauss. For the history, see Aldrich (1997), "R. A. Fisher and the making of maximum likelihood 1912–1922", *Statistical Science* 12: 162–176, and Stigler (2007), "The epic story of maximum likelihood", *Statistical Science* 22: 598–620.

Slide 6 — Probability and likelihood

The formula of the likelihood is the formula of a joint distribution from Chapter 2. What changes is what we hold fixed. In Chapter 2 the parameters were fixed and we calculated the probability of different possible samples, so the data varied. Now the sample has been observed, so the observations are fixed numbers, and we look at how the probability of that sample changes as the parameter varies. A parameter value under which the observed sample is more probable is better supported by the data. That is the crucial idea behind likelihood, and everything in the rest of the chapter, the estimates, their confidence intervals and the comparison of models, is calculated from it.

The Poisson mass $e^{-\lambda}\lambda^x/x!$ is one expression that we can read in two ways. For a fixed λ it is a function of x: the probability of each possible count. For an observed x it is a function of λ: the likelihood of each possible mean.

Delivery

Allow one minute for the columns and one for the example. Write the expression once and point to x, then to λ, saying which of the two is fixed in each reading.

Slide 7 — Probability mass and likelihood curve

The two figures show the two readings of slide 6 for the fir seedlings, so that students have a picture of each before we calculate with them. The left figure is a probability mass function, as in Chapter 2: the mean is fixed at 2 seedlings per quadrat and the horizontal axis holds the possible

counts in one quadrat. Each height is the probability of a count (about 0.14 for an empty quadrat and 0.27 for one and for two seedlings), and the heights add up to one.

The right figure holds the 50 observed counts fixed and puts the mean on the horizontal axis. Its height at each value of λ is the probability of the whole observed sample under a Poisson model with that mean: the product of 50 probabilities like those on the left, one at each observed count. The dashed line marks $\lambda = 2$, the model of the left figure, where the likelihood is about 1.3×10^{-39}. The curve is highest at about 2.14, the mean of the counts, which slide 14 names as the estimate.

Figure walkthrough

Point at the horizontal axes first: possible counts on the left, possible means on the right. Then point at the vertical axis on the right, in units of 10^{-39}: the values are tiny, because they are probabilities of a sample of 50 counts, and they are compared between values of λ rather than read one at a time. The figure cannot show that the area under the curve is not one; slide 9 gives that number. The calculation of the likelihood at a given λ is in the chapter section "The likelihood function", and slide 15 draws the same kind of curve on the log scale. Included in the two minutes.

Slide 8 — A likelihood by hand

Each observation contributes its probability, and the contributions are multiplied. To compare two parameter values, we compare their likelihoods: the ratio says how many times more probable the sample is under one value than under the other. The value of a likelihood has no meaning by itself. Every observation multiplies it by a probability smaller than one, so its value depends on the number of observations, and only comparisons between parameter values for the same data are used.

The example takes the first three quadrats, which contain 0, 1 and 2 seedlings. The three contributions are $e^{-\lambda}\lambda^0/0!$, $e^{-\lambda}\lambda^1/1!$ and $e^{-\lambda}\lambda^2/2!$, with $0! = 1$, and their product is $e^{-3\lambda}\lambda^3/2$. At $\lambda = 1$ it is $e^{-3}/2 \approx 0.0249$ and at $\lambda = 2$ it is $e^{-6}\,2^3/2 \approx 0.0099$, so these three counts are about 2.5 times more probable under a mean of 1 than under a mean of 2. That is not surprising, because their own mean is 1.

The comparison depends on the data. For all 50 quadrats the likelihood is 4.2×10^{-50} at $\lambda = 1$ and 1.3×10^{-39} at $\lambda = 2$, so the order is reversed: the sample of 50 counts is about 3×10^{10} times more probable under a mean of 2 than under a mean of 1. Fifty quadrats discriminate between the two values far more strongly than three, and the mean of the 50 counts is 2.14. Note how small both numbers are, which is the point of the second bullet.

Delivery

Do the first calculation on the board or by pointing along the slide, and stop after the comparison of 0.0249 and 0.0099. Mention the numbers of the 50 quadrats in speech only; they are not on the slide. These calculations are all in the chapter, where `dpois()` and `prod()` verify them.

Slide 9 — A likelihood is not the probability of a parameter value

A likelihood is calculated from probabilities, but it is not a probability distribution for the parameter, for two reasons. First, we work with aleatoric probability (Chapter 2): probabilities describe how data or estimates vary across hypothetical samples, whereas the parameter of the model is a fixed quantity. In the model there is no variation about λ to which a probability could refer. Second, for a fixed λ the probabilities of all possible samples add up to one, but nothing forces the likelihoods of all possible values of λ to do so for a fixed sample. For the 50 counts, the area under the likelihood across all the values of λ is about 8.7×10^{-40}, very far from one.

So we use a likelihood to compare parameter values and never to say how probable one of them is.

Optional remark (Bayesian inference)

The likelihood is also at the core of Bayesian inference, where it is multiplied by a prior distribution and rescaled to give a probability distribution for the parameter. Chapter 1 explains why this book stays with the Frequentist interpretation. One sentence is enough here.

Slide 10 — Discussion: probability or likelihood?

This discussion checks the difference between the two readings before the next slide builds on it.

Discussion and expected answers — included in the three minutes

Give students one minute to decide with a neighbour, then take one answer per item.

1. **Probability:** the parameter is fixed at 2 and the count varies. It is a value of the mass function, $e^{-2}2^3/3! \approx 0.18$.
2. **Likelihood:** the counts are fixed and we compare two parameter values. This is the comparison of slide 8, with a ratio of about 2.5 in favour of $\lambda = 1$.
3. **Probability:** for a fixed λ the probabilities of all possible counts add up to one.
4. **Likelihood:** a curve over values of λ whose area is not one. The area of 8.7×10^{-40} is the one of slide 9.

Slide 11 — The log-likelihood

The **log-likelihood** is the natural logarithm of the likelihood, $\ell(\theta) = \log(L(\theta)) = \sum_{i=1}^{n} \log(P(X_i = x_i \mid \theta))$. It has three advantages. The logarithm turns the product into a sum, which is easier to work with. It is an increasing function, so a parameter value with a higher likelihood also has a higher log-likelihood and both functions have their maximum at the same value of θ. And a product of many probabilities can become smaller than the smallest positive number that the computer can represent, whereas a sum of their logarithms is an ordinary number. Once the data are fixed we write $L(\theta)$ and $\ell(\theta)$ for short.

For continuous data the contributions are densities. Recall from Chapter 2 that a density has units (one over the unit of the measurement), that it changes with the units and that it can be larger than one. The log-likelihood inherits these properties: it can be positive, and its value changes if we express the same measurements in other units. If body masses are converted from grams to kilograms, the density of every mass is multiplied by 1,000, because the same probability is spread over a thousand times fewer units, so the log-likelihood of n masses increases by $n \log(1000)$. This

is a constant, so the differences between the log-likelihoods of two parameter values, or of two models, are the same in both units. In general only differences between log-likelihoods calculated from the same data are meaningful. The practical checks this with body masses (Exercise 4).

The example simulates 1,000 counts from a Poisson distribution with $\lambda = 4$ (seed 501). R returns zero for `prod()` of their probabilities, although none of the probabilities is zero, while the sum of the log probabilities is -2094.6.

Slide 12 — Log-likelihoods in R

In R, all the `d*` functions have an argument `log`. With `log = TRUE` they return the logarithm of the probability mass (or density), which is more accurate than calculating the probability and taking its logarithm afterwards, and the log-likelihood is the sum of these values.

The chunk is the one of the chapter. `data()` loads the data set `fir` from the package `boot` without attaching the package, and the counts are its column `count`. The chunk before it in the chapter loads the packages RTMB and RTMBdist, RTMB first, as in Chapter 2: the distribution functions that we use come from them. The log-likelihood at $\lambda = 2$ is -89.52, and the second line, which takes the logarithm of the product, gives the same number. We can do that here because 50 probabilities still give a product that R can represent; with the 1,000 counts of the previous slide it would return minus infinity.

Delivery

Show the chunk without running it. If students ask about `getAll()` or `MakeADFun()`, say that they appear four slides later.

Slide 13 — Practice 1: crab burrows

Short activity — allow about three minutes of work and two to discuss

Students work on paper. Wait about three minutes and then go through the solution.

The likelihood is the product of the three contributions, $e^{-\lambda}\lambda^3/3! \times e^{-\lambda}\lambda^1/1! \times e^{-\lambda}\lambda^2/2! = e^{-3\lambda}\lambda^6/12$. For $\lambda = 2$ it is $64e^{-6}/12 \approx 0.0132$ and the log-likelihood is $-6 + \log(64/12) \approx -4.33$. For $\lambda = 3$ it is $729e^{-9}/12 \approx 0.0075$ and the log-likelihood is $-9 + \log(729/12) \approx -4.89$. The data are better supported by $\lambda = 2$, which is the mean of the three counts. The difference between the two log-likelihoods is 0.57, so the counts are about 1.8 times more probable ($e^{0.57}$) under $\lambda = 2$ than under $\lambda = 3$.

Common slips are to forget the factorials ($3!\,1!\,2! = 12$) and to compare the log-likelihoods as if they were likelihoods. Remind students that a difference of log-likelihoods is the logarithm of a ratio of likelihoods.

Break (10 minutes).

Slide 14 — The maximum likelihood estimator

The **maximum likelihood estimator** (MLE) of a parameter is the value for which the likelihood of the observed sample is highest, $\hat{\theta} = \arg\max_\theta L(\theta \mid x_1, \ldots, x_n)$, and it is the same as the value that maximises the log-likelihood. The symbol arg max reads "the value of θ at which the function

reaches its maximum": max would give the height of the highest point and arg max gives its location. In plain English, the MLE is the parameter value under which the model assigns the highest probability to the data that we observed. As in Chapter 3, the procedure of calculation is called the estimator and the value that it returns for a given sample is the estimate, and we use the abbreviation MLE for both.

At the highest point of the log-likelihood its slope is zero. We can differentiate the log-likelihood, set the derivative to zero and solve for the parameter. A solution that can be written as a formula is called a closed-form solution, and it exists only for a few models. For the others, the equations that we obtain cannot be solved with algebra, and we search for the maximum numerically. That is the method that we use in the rest of the book, because it works for any model for which we can write a likelihood.

For i.i.d. Poisson counts the closed-form solution is the mean of the sample, $\hat{\lambda} = \overline{x}$, which for the 50 counts is 2.14 seedlings per quadrat. It is also the method-of-moments estimator of Chapter 3. The derivation is in the chapter, in the box "Statistical detail: deriving the Poisson estimator", and the calculus refresher has the rules of differentiation. The Normal model is another case with a closed form: the MLE of μ is the sample mean and the MLE of σ^2 is the variance with divisor n, so the MLE of the variance is biased.

Delivery

Do not derive the Poisson estimator on the board. The derivative of the log-likelihood with respect to a parameter is called the score; name it only if a student asks.

Slide 15 — The likelihood curve

A **likelihood curve** is the log-likelihood of a model with one parameter, drawn as a function of that parameter. We draw it as the relative log-likelihood, $\ell(\theta) - \ell(\hat{\theta})$, so that the curve is zero at the estimate and negative everywhere else. All the likelihood figures of the chapter are on this scale, because only differences in log-likelihood are meaningful. How fast the curve falls on each side of the maximum tells us how well the data discriminate between parameter values, which is what Lecture 2 uses to calculate confidence intervals, and the curve need not be symmetric.

The example uses the first ten quadrats, because the shape of the curve is easier to see in a small sample; Lecture 2 compares ten quadrats with 50. They contain 22 seedlings, so the estimate is 2.2 seedlings per quadrat. The curve falls more steeply to the left of the maximum than to the right: the relative log-likelihood is -3.33 at $\lambda = 1.2$ and -1.76 at $\lambda = 3.2$, which are both one seedling away from the estimate, so a mean of 3.2 is better supported by these data than a mean of 1.2.

Figure walkthrough — included in the four minutes

The figure is a copy of the chapter figure. The horizontal axis is the mean number of seedlings per quadrat from 0.8 to 5 and the vertical axis is the relative log-likelihood, cut at -8. The vermillion curve is the relative log-likelihood and the dashed line marks the estimate at 2.2, where the curve is zero. Point to the two sides: the left side reaches -8 at about 0.8 and the right side reaches it only at about 4.7. The chapter draws the curve by evaluating the log-likelihood on a grid of values of λ with a `for` loop; the code is on the chapter page.

Slide 16 — Numerical optimisation

An **optimiser** is an algorithm that searches for the parameter values at which a function reaches its minimum. It starts from values that we provide, evaluates the function, and moves step by step towards lower values until it cannot improve any further. A grid shows us the curve, but to find the highest point of a log-likelihood for any model, with any number of parameters, we use an optimiser. In R we use `optim()`. Because it minimises, we do not give it the log-likelihood but the **negative log-likelihood** (NLL), $-\ell(\theta)$: the parameter value with the lowest NLL is the one with the highest likelihood.

Figure walkthrough

The schematic has no numbers. The vermillion curve is the NLL against a parameter, shaped like a bowl. The dark green arrows are the steps of the optimiser, from the starting value on the left down the curve to its lowest point, which is the estimate on the horizontal axis. The picture cannot show how the optimiser chooses each step; the method that we use, called BFGS, is explained in Chapter 8.

Slide 17 — The negative log-likelihood in R

In this book every NLL is written in the same form, which the package RTMB needs. It is a function with two arguments, both of them named lists: the first holds the parameters (`parms`) and the second the data (`data`). Its first line calls `getAll()`, which makes the elements of both lists available by their names, so that the rest of the function reads like the model. It returns minus the sum of the log probabilities. For the Poisson model the list of parameters has one element, `lambda`, and the list of data has one element, `counts`.

Delivery

Say explicitly that this is the form of every NLL in the course, and that the first argument is always the parameters and the second the data. The chunk only defines a function; nothing is calculated yet.

Slide 18 — From the negative log-likelihood to a fit

Three steps take us from the function to the estimate, and each has one name in code format. First, `cmb()` is a small helper that takes an NLL and a list of data and returns the NLL of those data as a function of the parameters only. This way we can calculate the same likelihood later on other data, as long as the variables of the list have the same names. Second, `MakeADFun()` processes the NLL so that the tools that we use in the rest of the course can work with it (the details are explained progressively through the course). It takes the function returned by `cmb()` and a list with the starting value of every parameter, and it returns an object that has three elements that we need: `par`, the starting values as a vector; `fn`, the NLL as a function of that vector; and `gr`, the gradients of the NLL, which are the slope that RTMB supplies so that the optimiser can move efficiently (Chapter 6 explains how, and Chapter 8 the method). Third, `optim()` minimises `fn` starting from `par` and using `gr`.

Figure walkthrough

The schematic has four boxes joined by arrows, from left to right: the NLL (a function of parameters and data), `cmb()` (which adds the data), `MakeADFun()` (which adds the starting values and returns `par`, `fn` and `gr`) and `optim()` (which returns `par`, `value` and `convergence`, the next slide). Follow the arrows once, box by box, and do not explain what happens inside `MakeADFun()`.

Slide 19 — A fit in R

These are the three chunks of the chapter, shortened. In step 1 we define `cmb()`. In step 2 we connect `poisson_nll_lambda` to the 50 counts and process it, with a starting value of one seedling per quadrat for `lambda`; `silent = TRUE` stops `MakeADFun()` from printing its progress. In step 3 `optim()` minimises `fn` from `par` with the slope `gr`; the argument `method` chooses the algorithm and we use `"BFGS"` in most cases because it is efficient. The same three steps apply to every model in the course. Only the NLL, the data and the starting values change.

Delivery

This is a slide to read slowly: take about a minute on each step. Say that any starting value from which the optimiser reaches the maximum will do, and that the chapter will use the method of moments for the owls only as one possible choice.

Slide 20 — The result of `optim()`

`optim()` returns a list whose most relevant elements are `par`, the value of the parameter at the minimum (the maximum likelihood estimate), `value`, the value of the NLL at that estimate, and `convergence`, a code that is zero when the algorithm stopped normally. Check `convergence` after every fit. If it is different from zero, the search did not finish normally and the values in `par` are not estimates. For the 50 counts `par` is 2.14 seedlings per quadrat, the mean of the sample, as the closed-form solution said.

Delivery

The chapter has a note callout "When a fit fails". Mention three things from it. While it searches, `optim()` can try values for which the NLL cannot be calculated, and `fn` then returns `NaN` without a warning; BFGS treats this as a failed step, so it is harmless when the search ends with convergence code 0 and sensible estimates. When the NLL is `NaN` at the starting values, `optim()` stops with the error "initial value in 'vmmin' is not finite". If a fit fails, check that every constrained parameter is transformed inside the NLL, try other starting values and calculate the NLL at the starting values, for example with `fn(par)` of the object. Chapter 8 explains why optimisers fail and what else can be done.

Slide 21 — Constraints and transformations

A **constraint** is a restriction on the values that a parameter can take. The mean of a Poisson distribution must be positive, and a probability must lie between zero and one. An optimiser does not know about these restrictions: it can try any real number, and for a negative λ the function `dpois()` returns `NaN`. In the fit above the search stayed among positive values, but that is not guaranteed.

We deal with constraints by transforming the parameter. The optimiser works with a quantity that can take any real value, and the NLL converts it into the parameter of the model. For a positive parameter we use the logarithm, so the optimiser searches over $\log(\lambda)$ and `exp()` turns it into a positive mean inside the NLL. For a probability we use the logit, $\text{logit}\,(p) = \log(p/(1-p))$, which covers all the real numbers as p goes from zero to one; `qlogis()` calculates it and `plogis()` is its inverse (the names come from the logistic distribution). In the chunk the only change is the line `lambda = exp(log_lambda)`, and the list of starting values now has `log_lambda`. The element `par` of the fit is the estimate of $\log(\lambda)$, so we transform it back with `exp()` to report 2.14 seedlings per quadrat.

Delivery

The same NLL can be connected to the first ten quadrats by changing only the list of data, which gives 2.2 seedlings per quadrat. This is the reason for `cmb()`, and both objects return in Lecture 2, so say that in one sentence and move on.

Slide 22 — Invariance

Fitting on a transformed scale and reporting on the original scale relies on a property of maximum likelihood called **invariance**: the MLE of a function of a parameter is that function of the MLE, $\widehat{g(\theta)} = g(\hat{\theta})$. The reason is that a transformation only relabels the horizontal axis of the likelihood curve. Every value of $\log(\lambda)$ corresponds to one value of λ with the same likelihood, so the highest point is the same on both scales. The optimiser found the MLE of $\log(\lambda)$, and its exponential is the MLE of λ.

Invariance also gives the MLE of anything calculated from the parameter. Under the Poisson model the probability that a quadrat has no seedlings is $e^{-\lambda}$, and its MLE is $e^{-\hat{\lambda}} = e^{-2.14} \approx 0.12$. In the sample, 7 of the 50 quadrats are empty, a fraction of 0.14. Invariance carries over the estimate but not unbiasedness: Chapter 3 showed that $\hat{\lambda}$ is an unbiased estimator of λ whereas $e^{-\hat{\lambda}}$ is a biased estimator of $e^{-\lambda}$.

Optional remark (citation)

Zehna (1966), "Invariance of maximum likelihood estimators", *Annals of Mathematical Statistics* 37: 744, gives the general statement.

Slide 23 — Practice 2: back to the ecological scale

Short activity — allow about two minutes of work and one to discuss

The colleague's two models are fitted with `optim()`. Students use R as a calculator, or the calculator of their phone.

We apply the inverse of each transformation: `exp(0.92)` gives 2.5 ticks per lizard and `plogis(q = -0.85)` gives a germination probability of 0.30. By invariance they are the maximum likelihood estimates of the mean and of the probability, because the optimiser found the maximum likelihood estimates of the logarithm and of the logit. Ask what would be wrong with reporting `par` directly: it is on the scale of the optimiser, where a mean of 0.92 ticks and a probability of -0.85 would not make sense. The break follows.

Break (10 minutes).

Slide 24 — Models with several parameters

With several parameters the likelihood is a function of all of them jointly, and the MLE is the combination of values with the highest likelihood. The definitions of the previous slides do not change, but the likelihood is now a surface over the parameters and not a curve. This raises questions that did not exist with one parameter: how to look at the likelihood, how the estimates of the different parameters depend on each other, and what the data say about one of them. They matter for the rest of the book, because the models of Chapters 6 and 7 always have several parameters.

The example is the barn owl data of the Chapter 3 practical. The data set `Owls` of the package `glmmTMB` contains observations of barn owl nests (Roulin and Bersier 2007). For each visit of a parent to a nest, the variable `SiblingNegotiation` is the number of calls that the nestlings made. As in the practical, we keep the visits of the female parent to nests where the nestlings had been fed (`Satiated`), so that all the counts can reasonably share one mean. There are 122 visits, from 24 nests, with a mean of 4.75 calls per visit and a variance of 42.3, almost nine times the mean.

Because of this overdispersion we use a Negative Binomial model with mean μ and shape k, with $\text{Var}\ (X_i) = \mu + \mu^2/k$ (the supplement describes the distribution). We assume that the visits are i.i.d., which is a simplification because visits to the same nest may resemble each other; Chapter 7 covers models for such data.

Slide 25 — A fit with two parameters in R

The NLL is written as before. The list of parameters now has one element for each parameter, `log_mu` and `log_k`, and both parameters are positive, so we transform both with the logarithm. The list of data has one element, `calls`. For the probability mass we use `dnbinom2()` of RTMBdist, which is written in terms of the mean (`mu`) and the shape (`size`). The optimiser needs a starting value for each parameter. The method-of-moments estimates, calculated as in the Chapter 3 practical ($\hat{\mu} \approx 4.75$ and $\hat{k} \approx 0.60$), are a sensible choice because they are quick to calculate and estimate the same parameters, but any values from which the optimiser reaches the maximum would do. `optim()` keeps the names of the parameters in its result, so we can select each estimate by its name and transform it back.

The estimate of the mean is the mean of the sample, 4.75 calls per visit: for a Negative Binomial model the MLE of the mean is the sample mean, whatever the value of k. The estimate of the shape is $\hat{k} \approx 0.33$, far from the method-of-moments estimate of 0.60 at which the search started. The two methods are different estimators, so they give different estimates from the same sample, and Lecture 2 compares their precision.

Delivery

The chapter has a warning callout on the two functions called `dnbinom2()`. RTMB and RTMBdist both have one, with different arguments (`mu` and `size` in RTMBdist, `mu` and `var` in RTMB). R uses the one of the package attached last, so always load RTMB first and RTMBdist second and name

the arguments. If only RTMB is loaded, R stops with an error about an unused argument `size`. Say this once and keep going.

Slide 26 — Checking the fitted model

`optim()` returns estimates for any model that we give it, so after a fit we compare the fitted model with the data. For counts we compare the relative frequency of each count in the sample with the probability that the fitted model assigns to it. As a reference we add a simpler model, a Poisson model with the same mean.

Figure walkthrough — included in the three minutes

The figure is a copy of the chapter figure. The grey bars are the relative frequencies of the number of calls per visit, from 0 to about 26. The vermillion line with points is the fitted Negative Binomial model and the dark grey line with points is the Poisson model. The bar at zero calls is 0.44 (54 of the 122 visits), and the Negative Binomial model gives it a probability of 0.41, whereas the Poisson model gives about 0.009. The Negative Binomial model spreads the rest of its probability over a wide range of counts, and the Poisson model concentrates it in a narrow peak around 4 and 5 calls. The log-likelihood of the Negative Binomial model is about 335 units higher (-299.44 against -634.16). Lecture 2 compares the two models with a formal criterion, the Akaike information criterion. The figure does not show that the Negative Binomial model is adequate; it shows that it is much closer to the data than the reference.

Slide 27 — The likelihood surface

With two parameters we can still evaluate the log-likelihood on a grid, which now has two dimensions: one for the values of μ and one for the values of k. The chapter stores the results in a matrix with one row per value of μ and one column per value of k, and fills it with two nested `for` loops. Inside the loops it calls `fn` of the processed object, which returns the NLL for a vector of parameters on the scale that was optimised, so we give it the logarithms of the candidate values and change the sign of the result. `contour()`, from the Chapter 1 practical, draws the matrix.

Each contour joins the combinations of μ and k with the same relative log-likelihood, like the contours of a joint density in Chapter 2, and the MLE is at the top of the hill. The contours are not tilted, which means that the estimates hardly depend on each other, because the best value of μ is the mean of the sample for every value of k. They are not symmetric either: the surface falls more slowly towards large values of μ and of k than towards small ones.

Figure walkthrough

The horizontal axis is the mean number of calls per visit (2.5 to 9) and the vertical axis is the shape (0.12 to 0.75). The grey lines are contours and the vermillion circle is the MLE. Point to the roughly upright egg-shaped inner contours and to the wider spacing towards the upper right.

Optional minute (interactive page)

If the connection allows, open Kristoffer Magnusson's page https://rpsychologist.com/likelihood/ for about a minute, inside the four minutes. It uses a Normal model with a mean and a variance and a sample of ten observations. In "Likelihood Calculation" move the sliders and watch how the

contribution of each observation and the log-likelihood change, and in "Finding the Maximum Likelihood Estimates" watch an algorithm climb the log-likelihood. The last part of the page is about hypothesis tests, which the course does not use. The static figure is the fallback.

Slide 28 — Parameterisation

The same model can be written with different sets of parameters, each of which is a **parameterisation**. By invariance, the maximum and the fitted model are the same in every parameterisation, but the shape of the surface around the maximum is not. The Negative Binomial distribution can also be written with the shape k and a probability $p = k/(k + \mu)$, which are the arguments `size` and `prob` of `dnbinom()`. In this parameterisation the maximum log-likelihood is -299.44, the same as before, and $\hat{p} \approx 0.064$.

Figure walkthrough — included in the four minutes

The figure is the chapter figure of slide 30, and here only the left panel matters. It has k on the horizontal axis and p on the vertical axis. The contours are elongated rings tilted upwards to the right: a larger k goes with a larger p, because the two together determine the mean, which the data pin down well. Say that the two lines and the right panel are explained on the next slides, and ask students to ignore them for now. Tilted contours describe estimates that depend on each other. Whether they are tilted depends on the model, its parameterisation and the data, so one example cannot be generalised.

The course writes distributions in terms of their mean where possible, because the models of later chapters describe how the mean changes with covariates. The choice of parameters also matters for numerical optimisation, which Chapter 8 covers.

Slide 29 — Slices and profiles

A surface can only be drawn for two parameters, and even then we usually want to know what the data say about one parameter at a time. There are two ways of reducing a likelihood surface to a curve for one parameter, and they answer different questions. Let θ_1 be the parameter of interest and θ_2 the other parameter, or parameters. A **slice** fixes the other parameters, normally at their estimates, and varies the parameter of interest: it is a cut through the surface along a straight line. A **profile** re-estimates the other parameters for every value of the parameter of interest, $\ell_{\text{prof}}(\theta_1) = \max_{\theta_2} \ell(\theta_1, \theta_2)$. In plain English, for each value of θ_1 we find the value of θ_2 that maximises the log-likelihood with θ_1 held fixed, and we record that maximum. Which parameter we profile depends on the question.

In the example, $\ell(k, \hat{p})$ is a slice and $\max_p \ell(k, p)$ is the profile of k.

Delivery

No name is introduced for the parameters that the profile re-estimates; say "the other parameters". The slide does not say how the profile is calculated. Slide 30 shows what it is and the demo shows how TMB calculates it.

Slide 30 — Slice against profile

The slice follows the horizontal line at $\hat{p}$, which leaves the high part of the surface quickly because the contours are tilted. The profile follows the ridge of the surface: for each k it moves to the best p. Both pass through the MLE, and everywhere else the profile is higher than the slice, because it is a maximum over all the values of p. The slice is therefore narrower. It describes what the data would say about k if we knew p exactly, whereas the profile takes into account that p is estimated from the same data. The profile does not depend on how the other parameters are written, whereas a slice does: with μ and k, where the contours are not tilted, the slice and the profile of k coincide.

We use profiles to calculate confidence intervals, in Lecture 2. Slices need no optimisation, so they are a quick way to inspect a fitted model, and they return as a diagnostic in Chapter 8.

Figure walkthrough — included in the four minutes

The figure is a copy of the chapter figure and has two panels. On the left, the tilted contours in k and p with a dashed vermillion line at $\hat{p}$ (the slice) and a solid vermillion line along the ridge (the profile), both passing through the circle of the estimate. On the right, the relative log-likelihood against k along the slice (dashed) and along the profile (solid). Both curves are zero at about 0.33 and the profile is above the slice everywhere else. The figure draws no cutoff. Within 1.92 units of the maximum, which is the level that Lecture 2 justifies and that the example of the slide uses to compare the two widths, the slice covers k from 0.26 to 0.41 and the profile from 0.24 to 0.45. Leave the questions about how the profile is calculated for the demo.

Slide 31 — Live demo: a likelihood profile in R

The demo shows on the owl model what slides 28 to 30 describe. It uses the script `demo-profile.R`, which runs in a clean R session in a few seconds and needs the packages RTMB, RTMBdist, TMB and `glmmTMB` (for the data). It writes the NLL in the parameterisation with k and p, so that the slice and the profile differ, and no profile is calculated by hand: TMB does it, as in the chapter. The script has a section "0. Packages and data" and then one section for each of the four steps of the slide.

Run section 0 during the second break, or while students arrive, and keep the R session open. It loads RTMB first and RTMBdist second, loads the owl data, calculates the method-of-moments estimates for the starting values and defines `cmb()`. Spend the two minutes at the slide reading the four steps, and then switch to R.

Walkthrough of the script — 16 minutes

1. **The fit (section "The fit with k and p", about 4 minutes).** The NLL `nb_prob_nll` has two parameters, `log_k` and `logit_p`: `k = exp(log_k)` and `p = plogis(logit_p)`, and `dnbinom()` with `size = k` and `prob = p`. Process it, starting from the method-of-moments estimates, and fit it. `convergence` is 0. The estimates are $\hat{k} \approx 0.327$ and $\hat{p} \approx 0.064$, and the maximum log-likelihood is -299.44, the same as with μ and k. Ask students why (invariance, slide 22 and slide 28). The surface of this model is the left panel of slides 28 and 30, calculated with the grid of `fn` of slide 27.

2. **The slice (section "The slice", about 3 minutes).** One `for` loop over a fine grid of k with p fixed at its estimate: each value needs only one call of `fn`, with `log(k)` and `qlogis(p_mle)`. Subtract the maximum log-likelihood to get the relative scale.
3. **The profile (section "The profile", about 5 minutes).** `TMB::tmbprofile(owl_prob_obj, name = "log_k", ytol = 6, trace = FALSE)`. Explain the arguments: the processed object, the name of the parameter as it is written in the list of parameters, and `trace = FALSE` to stop printing. The prefix `TMB::` is needed because RTMB does not export the function. At every value of `log_k` that it visits, `tmbprofile()` runs an inner fit that re-estimates `logit_p`, so this is the definition of slide 29 carried out by TMB. By default it stops about two units below the maximum, which is enough for a 95% interval; `ytol = 6` continues it to the lower limit of the plot. The data frame has the columns `log_k` and `value` (the NLL), which we convert to k and to the relative log-likelihood.
4. **Both curves (section "Both curves", about 3 minutes).** The plot shows the profile as a solid line and the slice as a dashed line, on the relative log-likelihood. The slice is narrower, and the profile is above it everywhere except at the maximum: the right panel of slide 30, drawn in base R. Ask the questions below.

Questions to ask during the demo (about 1 minute)

Ask "slice or profile?" for each item and take one answer:

1. k varies with p fixed at its estimate: a slice.
2. k varies with p re-estimated at each value: a profile.
3. What the data would say about k if we knew p exactly: a slice.
4. A curve that takes into account that p is estimated from the same data: a profile.

If time allows, use question 6 of the chapter quiz, "two curves for one parameter".

Slide 32 — The Chapter 4 R practical: first part

The practical has five exercises. After this lecture, students can do the first two partly. Exercise 1 is worked on the page, so read it first. It uses the survival of reed frog tadpoles in the 24 tanks with predators (`ReedfrogPred` of `emdbook`; Vonesh and Bolker 2005) under a Binomial model with one survival probability. Questions 1 to 3 apply this lecture: the likelihood contributions of the first three tanks by hand, the maximum likelihood estimator by differentiating the log-likelihood, and the NLL with the logit of p as the parameter, fitted with `optim()`. Question 3 also asks for standard errors, which belong to Lecture 2, so students leave them for later. Exercise 2 uses the number of glacier lily seedlings in 256 quadrats (`Lily_sum` of `emdbook`) under a Negative Binomial model, like the owls. Questions 1 to 3 are the method of moments, the fit and the contours of the surface; the joint confidence region of question 3 belongs to Lecture 2.

Try each question before opening the solution.

Slide 33 — Summary and closing questions

Go back to slide 3 by speaking about it again; no new slide is needed. The likelihood is the probability of the sample, read as a function of the parameters, and it is used to compare parameter values. The log-likelihood is a sum, and only differences between log-likelihoods are meaningful.

The MLE is the value with the highest likelihood, and we find it with `optim()` after writing the NLL in the form of the course, connecting it to the data with `cmb()` and processing it with `MakeADFun()`. Transformations keep parameters in range and invariance carries the estimate back to the ecological scale. With several parameters the likelihood is a surface whose shape depends on the parameterisation, although the maximum does not. A slice fixes the other parameters and a profile re-estimates them, and `TMB::tmbprofile()` calculates profiles for us.

Return to this slide for the two closing questions.

Discussion and expected answers — allow about two minutes

Question 1: because a likelihood is the probability of the data for a given value of the parameter, and the parameter is a fixed quantity of the model; there is no probability that refers to it. Also, the likelihoods of all possible parameter values need not add up or integrate to one.

Question 2: because a slice treats the other parameters as known, at their estimates, whereas a profile re-estimates them for every value of the parameter of interest, and so it never falls below the slice. When the other parameters are poorly determined, or their estimates depend on the parameter of interest, the difference is large.

Final transition

After a short practical session, Lecture 2 asks how uncertain these estimates are: what the curvature of the log-likelihood tells us about standard errors, how likelihood profile intervals are calculated, and why maximum likelihood works.

Adjustments during delivery

The slides take 127 minutes and leave 3 minutes of slack, which comes last. The demo is kept whole; if it overruns, it takes the slack first. If discussion overruns, use the slack first, then shorten in this order:

- Slide 3 to one sentence (“this lecture is the likelihood and the optimiser”): recovers about half a minute.
- Slide 4 to the table of the two columns: about one minute.
- Slide 20 to its bullets, with the callout on failed fits left to the chapter: about one minute.
- Slide 26 to the figure alone, without the sentence on the Poisson model: about one minute.
- Slide 27 without the optional minute of Magnusson’s page: about one minute.
- In the demo, skip the slice code and show only its curve in the final plot: about three minutes.

Keep the difference between probability and likelihood (slides 6, 7 and 9), the steps of a fit (slides 17 to 20), and slice against profile (slides 29 and 30).

The breaks are placed between the blocks, after slides 13 and 25 (minutes 38 and 95). If a block runs over, shorten the block rather than the following break.

If time is left: code to run

No code is run live apart from the demo. Students can follow the chunks of the chapter page at home, by section: “The likelihood function” (`prod(dpois())` for the first three quadrats and

the 50 counts), “The log-likelihood” (the 1,000 simulated counts with seed 501), “The likelihood curve” (the grid and the `for` loop for the ten quadrats), “Numerical optimisation” and “Constraints and transformations” (the fits of the Poisson model), and “Models with several parameters” (the owl fit and the grid of nested `for` loops for the surface).